

Paper 1B practice 3

Which drag law? An experiment that picks a side

IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **20 marks**

Show your working.

A falling object reaches terminal velocity when the drag force balances its weight. Two models are proposed for the drag force F on a paper coffee filter falling through air:

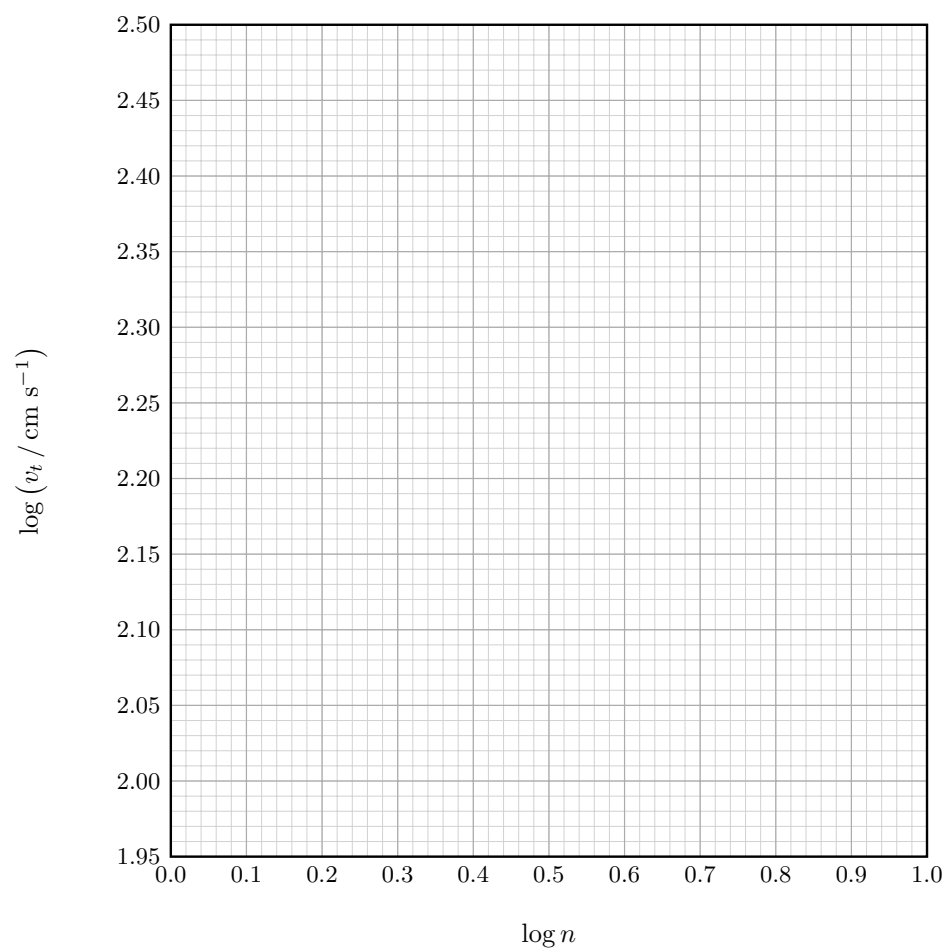
$$\text{model 1: } F = kv \qquad \text{model 2: } F = kv^2$$

where k depends only on the shape and size of the filter.

A student drops n identical filters nested together. Nesting keeps the shape and cross-sectional area unchanged, so k is the same for every stack, while the weight is n times that of a single filter. The terminal velocity v_t of each stack is measured.

n	$v_t / \text{cm s}^{-1}$	$\log n$	$\log(v_t / \text{cm s}^{-1})$
1	98	0.000	1.991
2	140	0.301	2.146
3	168	0.477	
4	197	0.602	2.294
6	205		2.312
8	278	0.903	2.444

- (a) Show that model 1 predicts $v_t \propto n$, while model 2 predicts $v_t \propto \sqrt{n}$. [3]
- (b) A graph of $\log v_t$ against $\log n$ is to be drawn. Explain why this distinguishes the two models, and state the gradient each one predicts. [3]
- (c) Complete the two gaps in the table. [2]
- (d) Plot the six points on the grid on the next page and draw the best-fit line, ignoring any anomalous reading. [3]
- (e) Identify the anomalous reading and justify leaving it out of the fit. [2]
- (f) Determine the gradient of your line. [2]
- (g) State which model the data support, and justify your choice using your gradient. [3]
- (h) Suggest one reason why a stack might fall more slowly than the model predicts. [2]



Answers

- (a) At terminal velocity the drag balances the weight, and the weight of a stack of n filters is nm_1g . *Model 1*: $kv_t = nm_1g$, so $v_t = \left(\frac{m_1g}{k}\right)n$, giving $v_t \propto n$. *Model 2*: $kv_t^2 = nm_1g$, so $v_t = \sqrt{\frac{m_1g}{k}}\sqrt{n}$, giving $v_t \propto \sqrt{n}$. Nesting is what makes this work: k is unchanged, so n is the only thing that varies.

- (b) Both models have the form $v_t = An^p$, and taking logarithms of both sides gives

$$\log v_t = \log A + p \log n.$$

So a graph of $\log v_t$ against $\log n$ is a straight line whose *gradient is the exponent* p , whatever A happens to be. Model 1 predicts a gradient of 1.0; model 2 predicts a gradient of 0.5. The two are far apart, so the experiment can decide between them rather than merely agreeing with one.

- (c) $\log 168 = 2.225$ and $\log 6 = 0.778$.
- (d) Five points lie close to a straight line; the point at (0.778, 2.312) sits noticeably below it. The best-fit line is drawn through the other five and the anomalous point is ignored.
- (e) The reading for $n = 6$. On the graph it lies about 0.07 log units below the line through the others, far outside the scatter of the remaining points, which sit within about 0.01 of it. A single reading that misses by roughly seven times the typical scatter is far more likely to be a mistake, a mistimed drop or a damaged stack, than evidence against the model, so it is excluded from the fit and the reason is recorded.
- (f) Using two well-separated points on the line, for example (0.000, 1.99) and (0.903, 2.44):

$$\text{gradient} = \frac{2.44 - 1.99}{0.903 - 0.000} = 0.50.$$

Accept 0.45 to 0.55. The gradient of a log-log graph has no unit.

- (g) The data support **model 2**, $F = kv^2$. The measured gradient of about 0.50 matches the exponent that model 2 predicts, and is nowhere near the 1.0 that model 1 requires. The difference between the two predictions is far larger than the scatter in the points, so the conclusion is safe rather than marginal. (Had the outlier been kept, the gradient would have come out near 0.48, which happens to leave the conclusion unchanged; that is luck, not method.)
- (h) Any one of the following, with a reason: the filters were released from too low a height and had not yet reached terminal velocity, so the average speed timed was below v_t ; the stack was crumpled or not nested cleanly, increasing its effective area and so its drag; draughts in the room opposed the fall; or the timing was started late, which lengthens the recorded time and lowers the calculated speed.